

SEDAR 30: Caribbean Blue Tang and Queen Triggerfish

Blue Tang:

Assessment Process

The ability to utilize length-frequency data is contingent upon having reliable estimates of life history parameters (von Bertalanffy parameters in particular). Studies on basic life history (e.g. age-growth relationships and estimating natural mortality) in the US Caribbean will greatly enhance the utility of the existing length-frequency data and should provide the greatest benefit to providing management advice in the short term. Studies should be carefully planned to ensure a representative sample of individuals by age/size, region, season etc. This type of research should be placed as a top priority for key species.

Blue tang are a seemingly fast growing fish species with a long life-span. Beyond the age of five, length information is not informative about the age-structure of blue tang populations. A better understanding about how fishing mortality influences population structure will come from collection of catch-at-age data. Sampling efforts should be carefully planned to ensure representative sampling of individuals by fishing gear, mode, region, season etc.

Fishery-independent surveys should be considered as a top research priority for additional data collection. Fishery-independent surveys designed using a rigorous statistical framework will allow for the collection of species-specific catch and effort data that can be used to develop indices of abundance. Indices of abundance are used in stock assessments to inform models about how a population may be changing over time. Fishery-independent surveys can also be used to supplement existing programs by collecting age, length, weight, and reproductive data.

It is essential that continued efforts to improve the data collection of fishery-dependent catch and effort statistics be made. More specifically, continued efforts to collect species-specific catch statistics will be important for future assessments.

Reviewer Recommendations

Cardinale Recommendations

The assessment team does provide an exhaustive shopping list for future data to be collected, which would greatly improve the capability of assessing the status of the Caribbean queen triggerfish and blue tang stock. However, I also suggest that effort should be devoted to selectivity experiments aimed to evaluate the theoretical changes in selectivity linked with the historical changes in the mesh size of the traps.

Chen Recommendations:

The AW panel recommends improving the quality of life history parameter estimates; developing a fishery-independent monitoring program; continuing the efforts to improve the collection of species-specific catch and effort data; and modifying the length-based total

mortality estimator to account for potential changes in selectivity. I consider these research areas are important for reducing the uncertainty and improving the quality of the assessment. The AW panel probably needs to prioritize the research recommendations and separate the short-term research plan from the long-term plan.

Given the problems associated with the data, an important research goal should be to improve the data quality and quantity. Short-term and long-term plans should be developed to achieve the goal. Short term research priority may include (1) improvement of life history data estimates and the quantification of their uncertainty in the form of probability distributions; (2) identification of major fishing areas and their spatio-temporal variability via conducting interviews with fishermen involved in the fishery; and (3) identification of potential approaches that can be used to estimate species-specific landing data (e.g., based on species composition of landings that become available in recent years). The long-term research plan should include the development of a fishery-independent monitoring program and continued improvement of the sampling protocol for the collection of fishery-dependent data (catch and effort).

Given the data limitations, I believe another research priority that should be addressed soon is to evaluate the performance of the length-based estimator (Gedamke and Hoenig 2006) for the total mortality. Based on the information available and with some assumptions, a queen triggerfish fishery can be simulated, following the approach used in Gedamke and Hoenig (2006). A simulation study can be conducted with this simulated fishery to evaluate the performance of this length-based estimator for estimating the total mortality. Different scenarios can be developed to identify key factors that may have significant impacts on the performance of the estimator. This can guide the future model development and data collection.

Trzcinski Recommendations:

I set this out more generally below. I cannot prioritize these well because I do not know 1) the species biology, ecosystem and fishery well, 2) the long-term assessment goals or 3) the financial constraints. I think the assessment team would benefit from a meeting to discuss these issues and help set out the overall assessment framework. But if prioritizing is at all useful given my limited knowledge, I would work on getting a fisheries independent survey together, I might even do this over the life history work although that should be done as well.

Presumably some discussion occurred about whether the method used was the best given the available data. I think it is important to review and recapitulate that argument in the introduction to the assessment report.

Queen Triggerfish:

Assessment Process

The ability to utilize length-frequency data is contingent upon having reliable estimates of life history parameters (von Bertalanffy parameters in particular). Studies on basic life history (e.g. age-growth relationships and estimating natural mortality) in the US Caribbean will greatly enhance the utility of the existing length-frequency data and should provide the greatest benefit to providing management advice in the short term. This should be placed as a top priority for key species.

Fishery-independent surveys should be considered as a top research priority for additional data collection. Fishery-independent surveys designed using a rigorous statistical framework will allow for the collection of species-specific catch and effort data that can be used to develop indices of abundance. Indices of abundance are used in stock assessments to inform models about how a population may be changing over time. Fishery-independent surveys can also be used to supplement existing programs by collecting age, length, weight, and reproductive data.

It is essential that continued efforts to improve the data collection of fishery-dependent catch and effort statistics be made. More specifically, continued efforts to collect species-specific catch statistics will be important for future assessments.

During the assessment workshop, the fishers from the USVI indicated that the selectivity pattern for queen triggerfish violated the assumption of knife-edge selectivity in the mean-length model. Efforts should be made to expand this model to accommodate other selectivity patterns.

Reviewer Recommendations

Cardinale Recommendations

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Chen Recommendations:

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